

and flexibility for different production jobs.

Collaborative robotics. The emergence of collaborative robots (cobots) has introduced a new dimension to PnP technology, enabling machines to work alongside humans safely. This collaboration opens up new possibilities for custom packaging and small-batch assembly. The UR5e by Universal Robots excels in various pick and place tasks due to its medium payload and long reach, offering flexibility and precision. Techman Robot's TM5, featuring integrated vision and easy programming, is ideal for object recognition and sorting in similar tasks. ABB's YuMi, targeting small parts assembly, has dual arms that operate both together and separately, ensuring high precision and safe human interaction.

Enhanced connectivity and IoT integration. The integration of IoT technology allows PnP machines to communicate with other machines and systems, improving process integration and data analytics. There's a trend towards using open source software like OpenPNP to retrofit existing PnP machines. This approach offers greater flexibility and better software support than proprietary systems. However, this can sometimes come at



Maxim SMT SIPLACE SX1/SX2 (Credit: <https://smt.asmpt.com/en>)

the cost of reduced speed or precision, which highlights the importance of optimising software settings for each specific machine.

Sustainability. With a focus on sustainability, newer PnP machines are designed to be more energy-efficient and reduce waste, aligning with global environmental goals.

Redesign of component feeders. Feeder design innovations enhance PnP machines' efficiency and precision. For instance, redesigning tape feeders to fit more tightly and lock solidly has been a focus. This includes using flexures to improve the feeder's functionality, especially when 3D-printed. The Maxim SMT machine has feeder capacity SIPLACE SX1/ SX2: 120 slots with 8mm X feeders.

The Panasonic AM100 complements its advanced nozzle technology as the machine is equipped with four feeder carts capable of accommodating up to 160 feeders. This high feeder capacity ensures that the AM100 can handle various component types and sizes, making it an adaptable solution for various electronic manufacturing needs.

The evolution of pick-and-place machines is a testament to the relentless pursuit of efficiency and precision in manufacturing. As we look to the future, it is clear that the integration of advanced technologies like AI, ML, and

IoT will continue to shape the landscape of industrial production. The convergence of hardware and software advancements enhances production capabilities and paves the way for a future where intelligent factories are the

norm. This journey into the future of manufacturing is not just about adopting new technologies but re-imagining what can be achieved on the production floor. These advancements indicate a move towards more versatile, efficient, and user-friendly PnP machines capable of adapting to various production environments and requirements.

The advancements in pick-and-place machine technology signify a leap forward in manufacturing automation. High-speed operation, enhanced precision, AI integration, improved user interfaces, and robotic integration are just a few of the developments that have made these machines more efficient, flexible, and indispensable to modern production lines. Models such as the Yamaha YSM20R, ASM SIPLACE TX series, Juki RS-1R, Mycronic MY700, and Universal Robots UR3e exemplify these advancements, showcasing what is possible when cutting-edge technology is applied to solve traditional manufacturing challenges. As technology continues to evolve, we can expect further innovations in pick-and-place machines that will continue to push the boundaries of what is possible in automated manufacturing. **EFY**

This article incorporating significant contributions from Shailendra Mathur of MaximSMT and Shreyansh Gupta of DJK has been authored by Akanksha Sondhi Gaur, a technophile at EFY having a German patent. She has an interest in writing on subjects concerning electronics, and has authored an array of research papers



AM100 single-beam, single-head placement machine (Credit: <https://na.panasonic.com/us>)